

UCK 358E – Introduction to Artificial Intelligence

Final Project

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Aerospace Domain-Specialized Small Language Model Project:

This project will serve as the final examination of the course. Students are expected to work individually or in groups of up to five members to develop and evaluate a domain-specialized small language model for an aerospace engineering application. The project aims to combine concepts from artificial intelligence and aerospace engineering by adapting a local language model to perform as an expert assistant in a selected aerospace domain.

Each group must select one aerospace engineering specialization area such as aerodynamics, strength of materials, thermodynamics, automatic control, flight mechanics, propulsion, or a related field. The selected domain should be clearly defined, and all project components must focus on this area.

The mandatory base model for the project is Qwen3, specifically the Qwen3:1.7B version. Students must run the model locally and demonstrate its baseline performance before any modifications are applied. The project should investigate how the performance of a small LLM changes when different domain adaptation techniques are introduced. The core system must run locally. External cloud APIs or proprietary hosted AI systems may not replace the required implementation components.

The project consists of four main stages. First, students must evaluate the original base model in the selected aerospace domain. For this purpose, each team must prepare a domain-specific evaluation dataset and define suitable evaluation metrics. The evaluation dataset should contain at least 100 questions with varying difficulty levels, including conceptual engineering questions, analytical reasoning problems, numerical interpretation, engineering judgment, and equation-based questions. Students are expected to design appropriate metrics such as accuracy, engineering validity, completeness, hallucination rate, consistency, and retrieval quality.

In the second stage, students must prepare a domain-specific training dataset and fine-tune the model using parameter-efficient fine-tuning methods, specifically LoRA (Low-Rank Adaptation). The training data should contain at least 500 high-quality samples, although larger datasets are strongly encouraged. Students may use textbooks, academic publications, technical reports, lecture notes, NASA or FAA documentation, engineering manuals, and similar technical resources. The fine-tuning process, dataset preparation pipeline, and training methodology must be clearly documented. Simply collecting documents or PDFs is not sufficient; students are expected to demonstrate preprocessing, cleaning, chunking, annotation strategy, prompt formatting, or other meaningful dataset preparation steps.

In the third stage, students must develop a Retrieval-Augmented Generation (RAG) system using a knowledge base related to their aerospace domain. The knowledge base may include technical

documents, engineering equations, standards, definitions, material properties, or design principles. Students should implement a retrieval pipeline capable of retrieving relevant information and augmenting the model responses accordingly.

In the fourth stage, students must implement a GraphRAG system or a graph-based retrieval approach. This stage should demonstrate the use of structured relationships between engineering concepts. For example, relationships between lift, drag, angle of attack, stability, center of gravity, entropy, pressure, or control response may be represented as graph structures to improve retrieval and reasoning capabilities. Lightweight graph implementations are acceptable. Students may use tools such as simple entity-relation graphs or graph libraries instead of complex industrial-scale graph infrastructures. The emphasis is on engineering reasoning and retrieval methodology rather than software complexity.

The final objective of the project is to compare the performance of four different systems: the original base model, the fine-tuned model, the model enhanced with RAG, and the model enhanced with GraphRAG. Students are expected to perform a detailed comparative analysis discussing the advantages and limitations of each approach. The comparison should include quantitative results, qualitative analysis, hallucination behavior, engineering reasoning quality, computational requirements, inference speed, and retrieval effectiveness. Experimental results should be supported with tables, graphs, and example outputs.

Students are expected to focus on correctness, evaluation quality, and engineering understanding rather than excessive system complexity. A smaller but well-designed and properly evaluated system is preferable to an ambitious but incomplete implementation.

Each group must prepare a final project report with a maximum length of 20 pages excluding references and appendices. The report should clearly explain the selected aerospace domain, methodology, datasets, implementation details, experiments, evaluation metrics, results, discussion, and conclusions. The project report must also include a failure analysis section discussing situations where the models produce incorrect, incomplete, hallucinated, or physically invalid engineering responses. Students should analyze and compare failure cases across the baseline model, fine-tuned model, RAG system, and GraphRAG system.

In addition, each group must prepare a presentation and present their work during the final examination period. Demonstrations are encouraged whenever possible. Furthermore, all project-related materials must be uploaded together with the final report and presentation materials. These may include source codes, trained models, datasets, vector databases, retrieval or GraphRAG files, evaluation scripts, and other files necessary to reproduce the project results.

Although students may work in teams, grading will include both group and individual assessment components. Every team member must attend the final presentation and explain their own contributions to the project. Individual scores may differ among team members depending on their technical contribution, presentation quality, understanding of the implemented methods, and ability to answer questions during the verbal assessment. A student who cannot adequately explain their contribution or demonstrate sufficient understanding of the project may receive a lower score.

than other members of the same team. Furthermore, any student who does not attend the final presentation will automatically receive a grade of zero for the project.

The project will be evaluated based on technical implementation quality, correctness of methodology, quality of experimental evaluation, report quality, presentation performance, and individual understanding of the work. Students are encouraged to use modern open-source AI tools and frameworks for fine-tuning, retrieval systems, vector databases, and graph-based reasoning systems.